MSBF
Merchant Sales-Based Financing Platform

MERCHANT CREDIT INTELLIGENCE

ENTERPRISE ARCHITECTURE

Enterprise Merchant Sales-Based Financing Platform

From merchant sales to governed funding decisions—and the intelligence to manage what follows.

750
Synthetic merchants
Two matched scenarios • 1,500 application-scenario paths

6,519
Reportable structures
Governed search before a no-feasible conclusion

13.7% / 3.6%
Final-offer authorization rate
Baseline 103/750 • Recession Energy 27/750

15,600
Monitoring account-days
130 funded scenario paths × 120 days

THE ARCHITECTURE • SEVEN FUNCTIONAL BANDS

Primary evidence progression → Bands are not module numbers

0 — Governance & run control (G0 · G1 · all stages)

- G0 — **Physical data foundation**: Schemas, keys, reference policy and typed source structures.
- G1 / M1.1 — **Governed run control**: Run identity, frozen parameters, seed, source and scenario snapshots.
- ALL STAGES — **Validation & acceptance**: Preflight, positive/negative controls, reconciliation and evidence.

1 — Merchant & daily foundation (M1.2–M1.6)

- M1.2–M1.3 — **Merchant + requested terms**: 750 merchants and applications; amount, remittance, payback and horizon.
- M1.4–M1.5 — **POS, settlement & deposits**: Daily sales, cash movements and liquidity; raw evidence stays distinct.
- M1.6 — **Matched scenario panel**: Baseline and Recession Energy preserve merchant and application identity.

2 — Evidence, features & capacity (M1.7–M1.12)

- M1.7–M1.8 — **Quality, verification & continuity**: Separate usable evidence, identity/fraud signals and operating continuity.
- M1.9–M1.10 — **Cash flow & merchant capacity**: As-of features, obligations, residual cash flow and post-financing liquidity.
- M1.11–M1.12 — **Resilience & comparative risk**: Operating archetypes and risk components; transparent synthetic proxy.

3 — Economics & G2 contracts (M1.13–M1.17)

- M1.13–M1.14 — **Exposure, loss & economics**: Path-weighted exposure, recovery, comparative loss, costs and contribution.
- M1.15–M1.16 — **Two consumption contracts**: Scenario-aware application evidence + acquisition attribution/cost companion.
- M1.17 / G2 — **Accepted evidence foundation**: Read-only 1,500-path interface; contracts, archives and source identity assured.

4 — Decision & counter-offer strategy (M2.1–M2.3)

- M2.1 — **Eligibility & policy routing**: Preserve evidence limits and hard stops; route decisionable, remediable paths.
- M2.2 — **Governed structure search**: 6,519 reportable structures; amount, remittance, payback and 30/60/90 days.
- M2.3 — **Final decision & authority**: Seven outcomes. Authorized counter-offers remain distinct from review.

5 — Funded lifecycle & servicing (M2.4–M2.9)

- M2.4–M2.5 — **Fund, activate & monitor**: 130 funded paths; 15,600 account-days. Seasoned pace + early hard triggers.
- M2.6–M2.8 — **Intervene, set up & execute**: Evidence-led review routing, permitted setup and synthetic servicing events.
- M2.9 — **Reconcile & certify accounts**: Events, returns/retries, remaining balances and permitted account states.

6 — Portfolio & certified consumption (M2.10–M2.12 · Power BI)

- M2.10 — **Portfolio performance endpoint**: Source-owned exposure, state, servicing and comparative portfolio KPIs.
- M2.11 — **Bounded terms & account simulation**: Offer/terms and account-treatment comparison across eight profiles.
- M2.12 → P01–P04 — **G3-v2 → Power BI**: 70/70 source controls PASS. Analytical views consume certified source facts.

DELIVERY & STRATEGIC HORIZONS

CURRENT TECHNICAL DELIVERY

- **Module 1 / G2** — Accepted evidence foundation
- **Module 2 / R26** — Accepted technical successor
- **Power BI R3** — P01–P04 complete

PREPARATION & DEVELOPMENT

Campaign Scale: 750 replay → 2,500 → 25,000

Module 3 development • separate acceptance and execution gates

CONCEPTUAL MATURITY HORIZONS

YEAR 1 — Control & learn
Define appetite, test evidence and decisions, observe account behavior.
Build a disciplined foundation.

YEAR 3 — Refine & compare
Extend cohort/vintage depth; compare scenarios, segments and allocation.
Scale only through governed gates.

YEAR 5+ — Enterprise intelligence
Forecasting, scenario planning, relationship insight and governed learning across portfolios.
Broader insight. Better-connected decisions.

FUTURE MODULE CONCEPTS

M4 Platform & Governance • M5 Advanced Intelligence
M6 Strategy Laboratory

Source & parameter identity
Frozen inputs; explicit run lineage.

Positive + negative controls
Expected behavior and safe failure.

Exact reconciliation
Keys, counts, events and balances.

Consumption contracts
Stage-owned facts; explicit boundaries.

GOVERNED CONSUMPTION • EACH HANDOFF RETAINS ITS SOURCE

Module 1 / G2 — Accepted merchant evidence → **M2 / R26** — Decision + synthetic lifecycle → **G3-v2** — Certified source consumption → **Power BI P01–P04** — Executive analytical views → **Future consumers** — Separate contracts / authority

Evidence. Decisions. Accountability.
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v2.1 | ENTERPRISE ARCHITECTURE

Deterministic synthetic evidence • Non-production • No real funds

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MERCHANT CREDIT INTELLIGENCE

MODULE 1 • EVIDENCE ARCHITECTURE

Governed Evidence, Contract & Acceptance Lineage

Turn daily business activity into decision-ready evidence.

750
Synthetic merchants
One application per merchant

180 days
Baseline operating history
Daily POS, settlement and deposit evidence

2
Matched scenario alternatives
Baseline • Recession Energy

1,500
Integrated G2 contract rows
Application × scenario • read-only boundary

G0 PHYSICAL FOUNDATION — **G1 GOVERNED RUN CONTROL** — Frozen seed, parameters, run identity and source snapshots govern the entire chain.

Primary evidence progression → • Shared upstream inputs feed later stages; not every SQL join is drawn.

01 MERCHANT & DAILY FOUNDATION

Establish identity, requests and matched histories.

M1.2 Deterministic merchant population

INPUT	Governed seed + reference profiles
LOGIC	Generate merchant entities, industries and relationships.
OUTPUT	**750 merchant profiles; linked entity records**
ROLE	Identity for applications and daily histories.

M1.3 Application & requested structure

INPUT	Merchant + relationship evidence
LOGIC	Construct amount, remittance, total payback and horizon.
OUTPUT	**750 applications; one per merchant**
ROLE	Requested terms are not a financing decision.

M1.4 Daily POS & settlement history

INPUT	Merchant + operating parameters
LOGIC	Generate daily sales, seasonal patterns and settlement behavior.
OUTPUT	**Merchant × date; 180-day POS history**
ROLE	Raw operating evidence, not derived features.

M1.5 Daily deposits & liquidity

INPUT	Merchant + POS settlement histories
LOGIC	Model cash movements, balances, liquidity support and NSF events.
OUTPUT	**Merchant × date; deposit/liquidity history**
ROLE	Missing evidence is not favorable liquidity.

M1.6 Matched operating scenarios

INPUT	Accepted POS + deposit histories
LOGIC	Preserve Baseline; apply the Recession Energy overlay.
OUTPUT	**Merchant × date × scenario; two alternatives**
ROLE	Same merchants and dates; controlled comparison.

02 EVIDENCE, FEATURES & CAPACITY

Separate evidence usability from operating strength.

M1.7 Source quality & confidence

INPUT	Application + seven source families
LOGIC	Assess completeness, freshness, consistency and reconciliation.
OUTPUT	**Application × source family; 5,250 quality records**
ROLE	Evidence confidence remains separate from credit risk.

M1.8 Verification, fraud & continuity

INPUT	Source-quality, identity and operating evidence
LOGIC	Evaluate verification, fraud and processor-continuity checks.
OUTPUT	**4,500 atomic checks + 750 application summaries**
ROLE	Separate evidence dimensions; preserve review and blocked paths.

M1.9 As-of cash-flow features

INPUT	Matched histories + quality/verification evidence
LOGIC	Compute trailing operating features at the decision-time cutoff.
OUTPUT	**1,500 wide snapshots + 54,000 long-form feature values**
ROLE	Source hashes and usability accompany application/scenario features.

M1.10 Obligations & residual capacity

INPUT	Requested terms + features + obligation evidence
LOGIC	Assess payment coverage, residual cash flow and post-financing liquidity.
OUTPUT	**Application × scenario; capacity and affordability evidence**
ROLE	Sales are not free cash flow. Preserve evidence and review gates.

03 RESILIENCE, RISK & ECONOMICS

Connect affordability, comparative risk and economic viability.

M1.11 Operating resilience

INPUT	Cash-flow features + capacity snapshots
LOGIC	Build operating archetypes and components; enforce matched stress floors.
OUTPUT	**1,500 resilience snapshots + 7,500 component records**
ROLE	Explain revenue, liquidity, burden, continuity and confidence.

M1.12 Integrated comparative risk

INPUT	Verification + capacity + resilience evidence
LOGIC	Persist seven risk components, weighted score, reasons and floors.
OUTPUT	**1,500 risk snapshots + 10,500 component records**
ROLE	Synthetic comparative proxy; not a calibrated probability of default.

M1.13 Exposure, recovery & loss

INPUT	Request + capacity + risk + recovery evidence
LOGIC	Build exposure paths, timing weights and supported recovery assumptions.
OUTPUT	**Application/scenario snapshots + daily exposure paths**
ROLE	Explicit exposure denominator; comparative loss, not CECL.

M1.14 Unit economics & contribution

INPUT	Request/channel + capacity + comparative loss
LOGIC	Separate revenue, costs, loss, capital charge, contribution and hurdle.
OUTPUT	**1,500 economic snapshots + 21,000 component records**
ROLE	Conditional-if-booked economics; modeled contribution, not audited profit.

04 CONTRACTS & INDEPENDENT G2 ASSURANCE

Two distinct contracts → read-only integration → end-to-end acceptance

M1.15 Application consumption contract

`M1_APPLICATION_CONSUMPTION v1`

1,500 latest + 1,500 immutable archive rows

750 paired comparisons • scenario-aware evidence

Accepted M1.7–M1.14 outputs; preserve hashes and lineage.

M1.16 Acquisition attribution & cost companion

`M1_ACQUISITION_CONSUMPTION v1`

750 latest + 750 immutable archive rows

Application-level • scenario-invariant companion

Attribution, allocated cost and M1.14 overlap bridge; additive only.

M1.17 End-to-end QA & G2 acceptance

`G2_M1_CONTRACT = PASS`

Sources + contracts + archives + evidence

Read-only integration • independent reconciliation

Assure boundary, identity and nonregression; retain provenance.

`M1_G2_CONSUMPTION_BUNDLE v1` — **1,500-row integrated boundary → Module 2 decisioning → G3-v2 → Power BI analytical consumption**

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MODULE 2 • DECISION & LIFECYCLE ARCHITECTURE

Governed Decisioning, Operations, Portfolio & G3-v2 Lineage

Search the structure. Separate authority. Follow the account.

6,519
Reportable financing structures
Amount • remittance • payback • 30/60/90 days

130
Funded scenario paths
103 Baseline + 27 Recession Energy

$1,385,300 / $344,200
Synthetic funded amount
Baseline / Recession Energy • alternative scenarios

15,600
Monitoring account-days
120 days for each funded scenario path

STRUCTURE SEARCH → FINAL AUTHORITY

FUNDED LIFECYCLE → PERMITTED OPERATIONS

PORTFOLIO → CERTIFIED CONSUMPTION

M2.1 Eligibility & policy routing

INPUT
G2 application, acquisition and policy evidence.

LOGIC
Separate insufficient evidence and non-remediable stops from decisionable, structure-remediable paths.

OUTPUT / GRAIN
Decisionable routes + preserved evidence / hard-stop reasons.

BOUNDARY
Eligibility and routing; no final funding authority.

M2.2 Pricing & counter-offer search

INPUT
Routed request; capacity, risk and economics.

LOGIC
Test amount, remittance, payback and 30/60/90-day structures inside governed boundaries.

OUTPUT / GRAIN
6,519 reportable structures; selected alternative or no-feasible proof.

BOUNDARY
Construct and test. Final offer authority belongs to M2.3.

M2.3 Final decision & offer authority

INPUT
Gate results, tested structures and matched scenarios.

LOGIC
Choose authority-first; enforce strict/review frontiers and the governed stress floor.

OUTPUT / GRAIN
1,500 final scenario paths; seven distinct outcomes and applicable terms.

BOUNDARY
Authorize or require review. A reviewable structure is not approval.

M2.4 Booking, funding & activation

INPUT
Final authorization + selected financing structure.

LOGIC
Book, fund and activate only the authorized synthetic financing paths.

OUTPUT / GRAIN
130 funded scenario paths: 103 Baseline + 27 Recession Energy.

BOUNDARY
Funding / portfolio activation; not operational setup.

M2.5 Daily remittance & monitoring

INPUT
Funded lifecycle, daily sales and R26 monitoring policy.

LOGIC
Calculate remittance, balance and pace. Season pace signals; preserve early hard triggers.

OUTPUT / GRAIN
15,600 account-day rows; 120 days per funded scenario path.

BOUNDARY
Watch / adverse eligible at day 21/30; 3 consecutive eligible breach days.

M2.6 Intervention & servicing strategy

INPUT
Monitoring endpoint + recent account behavior.

LOGIC
Route no action, targeted outreach review or temporary remittance review from evidence.

OUTPUT / GRAIN
130 routes: 126 no action / 3 outreach review / 1 temporary review.

BOUNDARY
Recommendation and review routing; not execution.

M2.7 Operational account setup

INPUT
Intervention + governed execution conditions.

LOGIC
Resolve permitted setup, account instructions and any review hold.

OUTPUT / GRAIN
130 setup states with explicit permission and hold treatment.

BOUNDARY
Permitted operational setup remains distinct from funding.

M2.8 Servicing execution

INPUT
Setup state, payment instructions and balances.

LOGIC
Simulate permitted payments, returns and retries; reconcile corrected cent rounding.

OUTPUT / GRAIN
130 summaries + permitted synthetic event detail.

BOUNDARY
Execute only permitted instructions; retain payment lineage.

M2.9 Account-state certification

INPUT
Execution events + source balances and instructions.

LOGIC
Reconcile payments, returns/retries, remaining balances and permitted states.

OUTPUT / GRAIN
130 account certifications; source / execution / balance reconciliation.

BOUNDARY
Certify account state with exact-zero reconciliation.

M2.10 Portfolio performance

INPUT
Monitoring, execution and certified account state.

LOGIC
Assemble endpoint exposure, state, exceptions and source-owned portfolio KPIs.

OUTPUT / GRAIN
130 portfolio rows; $6,061.03 Baseline certified exposure.

BOUNDARY
The portfolio endpoint owns its measures and exposure basis.

M2.11 Bounded terms & account simulation

INPUT
1,500 paths; 6,519 structures; 130 account paths.

LOGIC
Compare bounded offer/terms and account-treatment alternatives across eight profiles.

OUTPUT / GRAIN
52,152 structure-profile; 12,000 application-profile; 1,040 account-profile rows.

BOUNDARY
Bounded simulation under the accepted authority boundary.

M2.12 Enterprise certification

INPUT
Accepted sources + simulation and validation evidence.

LOGIC
Bind source, simulation, control, report and consumption identities.

OUTPUT / GRAIN
70/70 G3-v2 source controls PASS; exact versioned consumption boundary.

BOUNDARY
Certify source consumption. Power BI explains the source facts.

Primary evidence progression → • Cross-stage source dependencies are retained; the diagram is not an execution script.

GOVERNED SEARCH BOUNDARIES

Minimum amount 40% of request • Maximum reduction 60% • 30/60/90-day horizons

Economic hurdle • Authority-first selection • Matched stress floor

SEVEN OUTCOMES. THREE AUTHORITY CLASSES.

1,500 application-scenario paths • not 1,500 distinct merchants

AUTHORIZED		REVIEW REQUIRED		DECLINE	
130		**250**		**1,120**	
As requested	115	Counter-offer	212	Insufficient evidence	234
Counter-offer	15	Manual underwriting	38	No feasible structure	652
				Policy hard stop	234

A supportable structure can still require review.
Only authorized outcomes enter synthetic funding.

COUNTER-OFFER ≠ AUTOMATIC APPROVAL

Synthetic merchant 098 • $5,200 requested in both alternatives

BASELINE
$5,200
As requested • authorized

RECESSION ENERGY
$2,500
Counter-offer • authorized

Matched alternatives for one merchant—not a time sequence.

THE AUTHORITY DISTINCTION • SYNTHETIC MERCHANT 082

$53,800 request → $31,200 selected

Baseline counter-offer • REVIEW REQUIRED, not funded

RECONCILE THE LIFECYCLE, NOT JUST THE DECISION

Certified account endpoint • 130 funded scenario paths

126	3	1
Paid off	Review holds	Active, reconciled

BASELINE EXPOSURE BRIDGE

$7,823.28 − $1,762.25 = $6,061.03

M2.25 attention exposure − processed = M2.10 certified exposure

Zero unresolved exceptions ≠ zero remaining exposure.

G2 foundation preserved → R26 decision + synthetic lifecycle → G3-v2 source certification → Power BI P01–P04

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v2.1 | MODULE 2 LINEAGE
Deterministic synthetic evidence • Non-production • No real funds